



2 Real Analysis 2023

Tutorial 3

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Cardinality, countable sets

Problem 7 Show that the following sets are countable.

- (a) The set of negative integers.
- (b) The set of non-negative integers that are divisible by 3.
- (c) The set $\mathbb{Q}^2$.

Problem 8 Prove or disprove:

- (a) The union of three countable sets is countable.
- (b) The intersection of three countable sets is countable.

Problem 9 Let $a, b \in \mathbb{R}$, $a < b$. Show that the intervals $(0, 1)$ and (a, b) have the same cardinality. (Hint: Construct a linear bijection.)

Limits of sequences

Problem 10 (a) Prove by definition that the sequence $\left(\frac{1}{\sqrt{n}}\right)_{n=1}^{\infty}$ converges to the limit 0.

- (b) Prove by definition that the sequence $(0, 1, 2, 0, 1, 2, 0, 1, 2, \dots)$ diverges.

Problem 11 For the following sequences $(a_n)_{n=1}^{\infty}$, determine the number $N \in \mathbb{N}$ with the property that

$$|a_n - a| < \varepsilon \quad \text{for all } n \geq N,$$

if $\varepsilon = \frac{1}{10}$ and if $\varepsilon = \frac{1}{100}$:

$$(a) \ a_n = \frac{n+2}{n}, \quad (b) \ a_n = \left(\frac{1}{2}\right)^n.$$

Problem 12 Prove that $\lim_{n \rightarrow \infty} a_n = a$ if and only if $\lim_{n \rightarrow \infty} (a_n - a) = 0$.